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Cross-lingual distillation for domain knowledge transfer with sentence transformers

Ruben Piperno [a,b,1], Luca Bacco [a,c,1], Felice Dell'Orletta [a], Mario Merone [b,*], Leandro Pecchia [b,d]

[a] ItaliaNLP Lab, Institute of Computational Linguistics "Antonio Zampolli", National Research Council, Via Giuseppe Moruzzi, 1, Pisa, 56124, Italy
[b] Research Unit of Intelligent Technology for Health and Wellbeing, Department of Engineering, Università Campus Bio-Medico di Roma, Via Alvaro del Portillo, 21, Rome, 00128, Italy
[c] Research Unit of Computer Systems and Bioinformatics, Department of Engineering, Università Campus Bio-Medico di Roma, Via Alvaro del Portillo, 21, Rome, 00128, Italy
[d] Fondazione Policlinico Universitario Campus Bio-Medico di Roma, Via Alvaro del Portillo, 200, Rome, 00128, Italy

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ABSTRACT

Recent advancements in Natural Language Processing (NLP) have substantially enhanced language understanding. However, non-English languages, especially in specialized and low-resource domains like biomedicine, remain largely underrepresented. Bridging this gap is essential for promoting inclusivity and expanding the global applicability of NLP technologies. This study presents a cross-lingual knowledge distillation framework that utilizes sentence transformers to improve domain-specific NLP capabilities in non-English languages. Specifically, the framework focuses on biomedical text classification tasks. By aligning sentence embeddings between a teacher model trained on English biomedical corpora and a multilingual student model, the proposed method effectively transfers both domain-specific and task-specific knowledge. This alignment allows the student model to efficiently process and adapt to biomedical texts in Spanish, French, and German, particularly in low-resource settings with limited tuning data. Extensive experiments with domain-adapted models like BioBERT and multilingual BERT with machine-translated text pairs demonstrate substantial performance improvements in downstream biomedical NLP tasks. The proposed framework proves highly effective in scenarios characterized by limited training data availability. The results highlight the scalability and effectiveness of this approach, facilitating the development of robust multilingual models tailored to the biomedical domain, thus advancing global accessibility and impact in biomedical NLP applications.

1. Introduction

Tackling the underrepresentation of non-English languages in Natural Language Processing (NLP) is crucial for achieving technological equity and inclusivity worldwide. The limited availability of linguistic resources for non-English languages severely impedes the performance and reliability of NLP models, exacerbating the technological divide between well-supported languages like English and underrepresented ones [1]. This challenge is particularly pronounced in the biomedical domain, where the lack of non-English data hinders the development of effective NLP models for global healthcare applications. The increasing complexity of cross-lingual tasks necessitates deeper Transformer models, as highlighted by Li et al. [2] through their innovative grouping skip connection mechanism to improve neural machine translation tasks". Existing research underscores that the absence of corpora, annotation tools, and linguistic resources for non-English languages diminishes the accuracy and utility of NLP systems in medical and healthcare contexts, restricting access to advanced information and technologies in many parts of the world [3].

To address these disparities and enhance the NLP models' performance on downstream tasks in less-represented languages (and domains), we propose a cross-lingual Knowledge Distillation (KD) approach employing Sentence Transformers [4]. Notably, this study does not aim to propose a novel knowledge distillation method or modify its theoretical underpinnings. Instead, it seeks to apply established KD strategies in a complex multilingual and domain-specific context. Our approach involves translating English biomedical texts into the target languages (Spanish, French, and German) using a state-of-the-art multilingual model. By employing KD with Sentence Transformers,

* Corresponding author.
E-mail address: m.merone@unicampus.it (M. Merone).
[1] The first two authors contributed equally to this work.

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we align the embeddings between a teacher model and a student model, ensuring that the student model learns semantically rich and contextually appropriate representations, thereby improving its ability to handle biomedical texts in multiple languages. This alignment leads to improved downstream performance, particularly in scenarios with limited data availability.

Our study demonstrates the effectiveness of this cross-lingual KD approach in enhancing the capabilities of biomedical NLP models, making advanced biomedical information accessible and usable in multiple languages. Compared to existing literature, we employed a novel loss function, the Multiple Negatives Ranking Loss (MNRL), which positively impacts the alignment of embeddings. We provide a comprehensive analysis to answer the following unexplored Research Questions (RQs):

- **RQ1:** In a scenario where no English data is available, *can we transfer the domain knowledge from a model pre-trained on the given domain?*
- **RQ2:** In a scenario where English data is available, *can we transfer the task knowledge from a model fine-tuned on the given task?*
- **RQ3:** In a scenario where no English domain-adapted model exists, *what are the benefits of distilling the task knowledge from a model fine-tuned starting from the same model as the student?*
- **RQ4:** *What is the impact of the amount of data used in fine-tuning on the final performance?*
- **RQ5:** *What are the differences in internal representations between the teacher and student models after the knowledge distillation process?*

To answer these questions, we conducted extensive experiments utilizing domain-adapted models like BioBERT [5] and multilingual BERT (mBERT) [6] as teachers, distilling knowledge into a multilingual student model (mBERT) in target languages, which has been shown to be an appropriate encoder for cross-linguistic transfer learning [7]. This distillation involved aligning the embedding representations between English and machine-translated biomedical texts. Specifically, we employed the MNRL function [8], which enhances the model's ability to distinguish correct query-response pairs (i.e., English and non-English) while minimizing similarities between incorrect pairs. By using BERT-based models for both teacher and student, specifically BioBERT and mBERT, we avoided biases associated with different model architectures and effectively transferred domain and task knowledge from pre-trained or fine-tuned teacher models.

Completed the distillation process, each student model was fine-tuned with different data sizes, i.e., at 10%, 25%, 50%, and 100% of the original non-English target languages' corpora. In addition, we performed the Representational Similarity Analysis (RSA) to assess the evolution of the student models during distillation, highlighting the similarities between the teacher and student models. This analysis provided deeper insights into the impact of our knowledge distillation approach on model performance.

2. Related works

The development of cross-lingual NLP models can be broadly classified into two primary approaches: vocabulary alignment and knowledge distillation. Vocabulary alignment entails aligning the embedding spaces of different languages to position semantically similar words closer together, thereby enhancing translation quality and cross-lingual understanding. For example, Maiorca et al. [9] explored semantic space alignment to improve translations by ensuring that analogous concepts across languages occupy proximate positions in the embedding space. This research paved the way for advancements in lexical modeling and vocabulary acquisition, as highlighted by Yamaguchi et al. [10], who stressed the importance of efficient vocabulary adaptation techniques for enhancing cross-lingual model performance. Jiang et al. [11] proposed a transfer learning approach based on lexical constraints to enhance low-resource machine translation. Their method leverages a beam search algorithm and dynamic vocabulary updates, demonstrating significant improvements in translation quality by effectively handling domain-specific terms and reducing training data requirements. Liang et al. [12] extended the MUSE framework [13] for aligning domain-preferential parts with limited isomorphism and thus building multilingual word embeddings for domain adaptation in low-resource languages.

A comparative study [14] on zero-shot cross-lingual sentiment analysis using modern multilingual Transformer-based models and linear transformations highlighted that while large Transformer-based models like XLM-R achieve superior performance, smaller models with linear transformations can offer a pragmatic alternative in real-world applications due to their lower training time and computational requirements. Likewise, Shen et al. [15] proposed an unsupervised multilingual machine translation framework leveraging pre-trained cross-lingual encoders. Their approach, XLM-DM, relies on bilingual dictionaries and target-side monolingual data, achieving robust performance in both bilingual and multilingual settings without the need for parallel corpora.

With the maturation of the field, researchers investigated data augmentation and other strategies to strengthen cross-lingual models. Ostendorff et al. [16] emphasized the advantages of linguistic data augmentation in developing more robust multilingual models, highlighting the pivotal role of training data diversity and richness in model efficacy. Buonocore et al. [17] focused on adapting transformer-based biomedical language models for less-resourced languages. By using neural machine translation and high-quality in-domain corpora, they created specialized biomedical language models that significantly improve the processing of medical texts in languages other than English, particularly Italian. Similarly, Downey et al. [18] conducted a comparative analysis of various cross-lingual knowledge transfer methods, providing insights into the most effective strategies for adapting multilingual vocabularies and improving model accuracy. In a complementary effort, Karpov et al. [19] demonstrated the benefits of knowledge transfer in multi-task, multi-lingual transformer models for various dialog tasks, highlighting improvements in model performance with cross-lingual training data. Huertas-Tato et al. [20] introduced the Siamese Inter-Lingual Transformer (SILT), which efficiently aligns multilingual embeddings for Natural Language Inference (NLI) tasks. SILT significantly reduces the number of trainable parameters and achieves state-of-the-art performance on common benchmarks by leveraging siamese pre-trained multilingual transformers with a matrix alignment method, showcasing an innovative approach to cross-lingual NLI.

Further advancements were seen with the use of multilingual pre-trained models. Chau et al. [21] demonstrated that significant improvements could be achieved using multilingual BERT with small corpora and treebanks, even without extensive data resources, by leveraging the inherent multilingual capabilities of the model. This was further extended introducing the Cross-Lingual BERT Transformation (CLBT) for zero-shot dependency parsing [22], enhancing cross-lingual representations by learning linear transformations from contextual word alignments. Ahmad et al. [23] proposed a deep transfer learning framework for emotion detection in low-resource languages, leveraging resources from a resource-rich language. Their method employs cross-lingual word embeddings to transfer knowledge between English and Hindi, resulting in significant performance gains for emotion classification in Hindi texts. This work highlights the potential of cross-lingual transfer learning to address data scarcity issues in emotion detection. Building on these insights, Pikuliak et al. [24] provided a comprehensive survey of cross-lingual learning methods, identifying and analyzing four types of cross-lingual transfer based on what is being transferred. Their survey highlights the increasing popularity and potential of cross-lingual learning, particularly in overcoming data scarcity for low-resource languages by transferring knowledge from resource-rich languages like English. This work serves as a valuable resource for understanding the various cross-lingual learning techniques and their applications in NLP.

Additionally, others proposed innovative methods for initializing multilingual embeddings, facilitating more effective cross-lingual transfer by ensuring better starting points for model training [25,26].

Conversely, knowledge distillation involves transferring knowledge from a high-performing teacher model to a student model, frequently utilizing both multilingual and monolingual models to enhance cross-lingual tasks. Pioneered by Reimers and Gurevych [27], KD demonstrated how multilingual embeddings could be created from monolingual models, leveraging cross-lingual knowledge transfer. This seminal work opened avenues for subsequent research into the applications of KD in multilingual settings. Others [28,29] extended this concept by employing multiple teacher models and cross-lingual distillation to improve text classification tasks, showcasing the adaptability of KD across different languages: Wang et al. [28] introduced a structure-level KD approach for multilingual sequence labeling, where the student model learns from multiple teacher models to improve its performance across different languages. Yang et al. [29], on the other hand, developed a zero-shot-aware training strategy in their multilingual distillation approach. Their model incorporates multiple language branches to improve both supervised and zero-shot performance, ensuring the student model can handle unseen languages effectively. Building on this, Nateras et al. [30] proposed a hybrid knowledge-transfer approach for cross-lingual event detection, leveraging a teacher-student framework with hierarchical sample selection, achieving state-of-the-art results on multiple languages. Huang et al. [31] validated these findings by examining KD's effectiveness in multilingual information retrieval, further confirming its advantages in cross-lingual applications. Subsequently, they proposed Optimal Transport Distillation (OPTICAL) for improving cross-lingual information retrieval on low-resource languages, demonstrating significant performance improvements with minimal training data [32].

Additional advancements include the development of mCLIP [33], a retrieval-efficient dual-stream multilingual vision-language model employing Triangle Cross-modal Knowledge Distillation (TriKD) to achieve state-of-the-art performance in multilingual image-text retrieval. Tran [34] contributed by efficiently transferring pre-trained language models from English to other languages using KD, demonstrating significant performance improvements even with limited computational resources. Additionally, [35] proposed a Reinforced Transformer with Cross-Lingual Distillation (RTCLD) approach for cross-lingual aspect sentiment classification, integrating target-sensitive adversarial learning to enhance performance across various languages. Chen et al. [36] explored cross-domain knowledge distillation for retrieval-based question answering systems, which complements cross-lingual applications by addressing domain-specific challenges and enhancing KD's effectiveness across heterogeneous domains.

Table 1 summarizes all the related works that apply vocabulary alignment or Knowledge Distillation. As shown, KD is a central theme in literature. Our work extends these recent advancements by integrating KD to facilitate both cross-lingual and cross-domain knowledge transfer. Unlike most of the works in literature, we do not apply KD during the fine-tuning process. Among the studies that employ KD, only Reimers and Gurevych [27] devote specific attention to producing multilingual embeddings from monolingual models through cross-lingual distillation. This distinctive methodology enables their approach to effectively align monolingual and multilingual representations. As a result, we focus our comparison on their work, since their foundational contributions to multilingual KD closely align with our objectives in cross-lingual and cross-domain knowledge transfer using sentence transformers. Inspired by their work, we use KD to align embeddings across different languages. Importantly, our intention is not to propose a novel KD technique or alter the theoretical foundations of knowledge distillation; rather, we aim to apply established KD methods to a challenging multilingual and domain-specific context. Additionally, by concentrating on transferring knowledge from the complex and specialized biomedical domain, our method aims to develop robust models that can comprehend and process medical texts in multiple languages.

Unlike prior approaches, we introduced a significant modification to the loss function, replacing traditional ones like Mean Squared Error (MSE) with the Multiple Negatives Ranking Loss. This adjustment enhances the ability to handle the specific challenges of multilingual biomedical knowledge transfer, leading to significant performance improvements. While other methods did not use automatically translated sentences within specific domains, making them less flexible for languages lacking parallel data, we utilized the mBART [37] model to address this issue. This ensures the quality and consistency of the machine-translated texts, offering a more adaptable solution. Moreover, we conducted an extensive study varying the number of training samples to analyze the impact on model performance. Unlike previous approaches, we performed a comprehensive analysis to answer the five research questions posed in our study. Lastly, we applied Representational Similarity Analysis to evaluate how well the student models' embeddings align with the teacher model after distillation, providing deeper insights into the structural changes induced by our method.

3. Materials and methods

3.1. Dataset

We utilized a publicly available dataset from Hugging Face, which includes medical and health-related texts in four languages: English, German, French, and Spanish.[2] This dataset is carefully curated to cover a wide range of medical topics, serving as a valuable resource for textual data. The dataset is composed of two distinct parts, catering to both professionals and patients. Each version is tailored to its target audience. On average, the samples contain approximately 1420 characters and are categorized into hierarchical topic labels, beginning with macro-categories. For each language, we identified these macro-categories, resulting in 82 classes for both French and German, 76 for Spanish, and 75 for English.

To comprehensively evaluate the model's effectiveness and generalizability, the dataset was divided into training and testing subsets. The training set comprises 81,300 samples, while the test set includes 27,100 samples. This split allows for a thorough evaluation of the model's performance on unseen data. Then, we performed a pre-processing phase on the provided data.

Data pre-processing. We began by analyzing the sample distribution across all classes, as depicted in Fig. 1(a) for the English language. We retained only those classes with a minimum of 400 samples that were common across all languages, resulting in 18 classes per language, as shown in Table A.4 in Appendix. Next, we examined the distribution of sample lengths in tokens for each language, as depicted in Fig. 1(b) for the English language. To achieve comprehensive representation and manage excessively lengthy samples, we applied a maximum token length of 512.

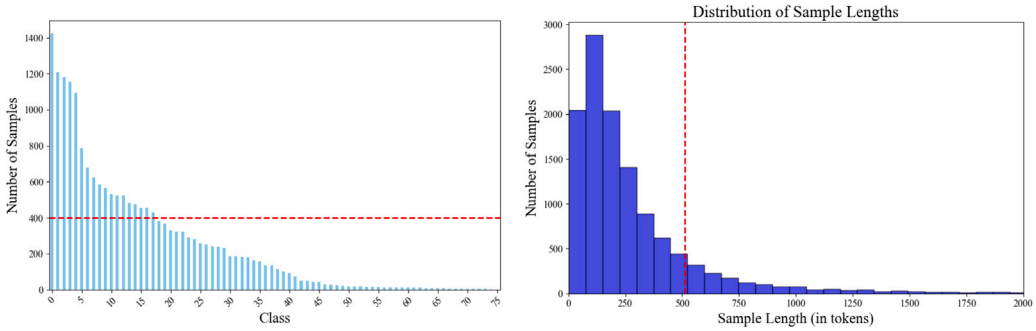
Following the preprocessing steps, the number of samples per class was reduced, as outlined in Table 2. We allocated 10% of the remaining training samples to the validation set. To implement our distillation method, we translated the English training set into the other languages using mBART [37]. mBART was chosen for its open-source availability, as well as its balance between strong performance and compact size compared to more advanced models.

To ensure that the test set did not include samples similar to those seen during training in the distillation process, we calculated the BLEU score between the English training set (translated into other languages) and the test set in other languages. This step was critical since all models were partially trained in English during the distillation process.

[2] Dataset available at: https://huggingface.co/datasets/nuvocare/MSD_manual_topics_user_base

Table 1
Summary of key works in multilingual NLP approaches, highlighting their task and domain.

Work	Year	Task	Domain
Vocabulary Alignment			
Wang et al. [22]	2019	Zero-shot dependency parsing	Cross-lingual Parsing
Chau et al. [21]	2020	Improvements using multilingual BERT	Dependency Parsing
Liang et al. [12]	2023	Multilingual word embeddings for domain adaptation	Low-resource Languages
Yamaguchi et al. [10]	2024	Efficient vocabulary adaptation	General NLP
Maiorca et al. [9]	2024	Semantic space alignment for translations	General NLP
Knowledge Distillation			
Reimers and Gurevych [27]	2020	Multilingual embeddings from monolingual models	Multilingual NLP
Wang et al. [28]	2020	Structure-level KD for sequence labeling	Sequence Labeling
Tran [34]	2020	Pre-trained models transfer using KD	General NLP
Chen et al. [36]	2021	Cross-domain KD for retrieval-based QA systems	Question Answering
Wu et al. [35]	2021	RTCLD for cross-lingual aspect sentiment classification	Sentiment Analysis
Jiang et al. [11]	2022	Transfer learning with lexical constraints	Machine Translation
Huertas-Tato et al. [20]	2022	Multilingual embeddings for Natural Language Inference (NLI)	NLI Tasks
Yang et al. [29]	2022	Zero-shot-aware training in multilingual distillation	Zero-Shot Multilingual NLP
Nateras et al. [30]	2023	Hybrid knowledge transfer for cross-lingual event detection	Event Detection
Huang et al. [31]	2023	Knowledge distillation for multilingual information retrieval	Information Retrieval
Huang et al. [32]	2023	OPTICAL for improving cross-lingual IR	Information Retrieval
Chen et al. [33]	2023	mCLIP using TriKD for multilingual image-text retrieval	Vision-Language Tasks



(a) Number of Samples per Class in English.

(b) Distribution of Samples' Length Expressed in Tokens.

Fig. 1. Dataset Information.

Table 2
Sample counts for each language.

Language	Train set	Validation set	Test set	Pre-Processed test set
English	11,860	1,325	4,295	–
Spanish	12,171	1,351	4,519	1,842
French	11,748	1,298	4,308	1,947
German	11,603	1,291	4,368	2,317

We excluded all samples presenting a maximum BLEU score higher than 0.3, as these scores are generally considered to indicate good similarity. Empirically, we observed that pairs in the dataset with scores above this threshold shared common meanings, while pairs below it presented different meanings. By setting this threshold, we retained only the most dissimilar samples compared to those seen during distillation, thus focusing on more challenging examples and excluding the presence of biases given by the KD process. Despite this filtering, we ensured that the distribution of the number of samples per class remained similar to that of the complete test set, as shown in Fig. 2. Table 2 also presents the remaining number of samples per language.

3.2. Proposed framework

Our framework is built on the fundamental principles of knowledge distillation through feature-level distillation [38]. Fig. 3 illustrates our framework. Similar to the work of [27], the aim of the methodology is to align the sentence embeddings between a teacher and a student model to create a new multilingual model. In this setup, the teacher model processes samples in the source language (English), while the student model processes both the source text and its translation into the target language (Spanish, French, or German). Unlike their approach, however, ours' objective is to transfer domain- and task-specific knowledge within the frame of low-resource languages and domains.

Furthermore, we substituted the traditional losses such as Mean Squared Error with the Multiple Negatives Ranking Loss function, which has shown to be particularly suitable for this kind of tasks in previous literature [8,39,40] and in the preliminary experiments of the present study. The MNRL function promotes the learning of embeddings that position correct pairs closer in the embedding space while separating incorrect pairs. It is defined as follows:

$$\text{MNRL} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp(sim(u_i, v_i)\cdot\alpha)}{\sum_{j=1}^{N}\exp(sim(u_i, v_j)\cdot\alpha)} \tag{1}$$

where N is the number of training samples (or pairs) in the batch, u_i is the embedding of the ith query (or sentence) in the batch, v_i is the embedding of the ith positive (or correct) response for the ith query in the batch, $sim(u_i, v_i)$ is the similarity function between the embeddings u_i and v_i, and α is a scaling factor applied to the similarity scores.

To handle the translation of training data, we leveraged transformer-based mBART [37], a pre-trained multilingual model with 50 languages and many-to-many translation capabilities.[3] This model represents one

[3] Model available at: https://huggingface.co/facebook/mbart-large-50-many-to-many-mmt

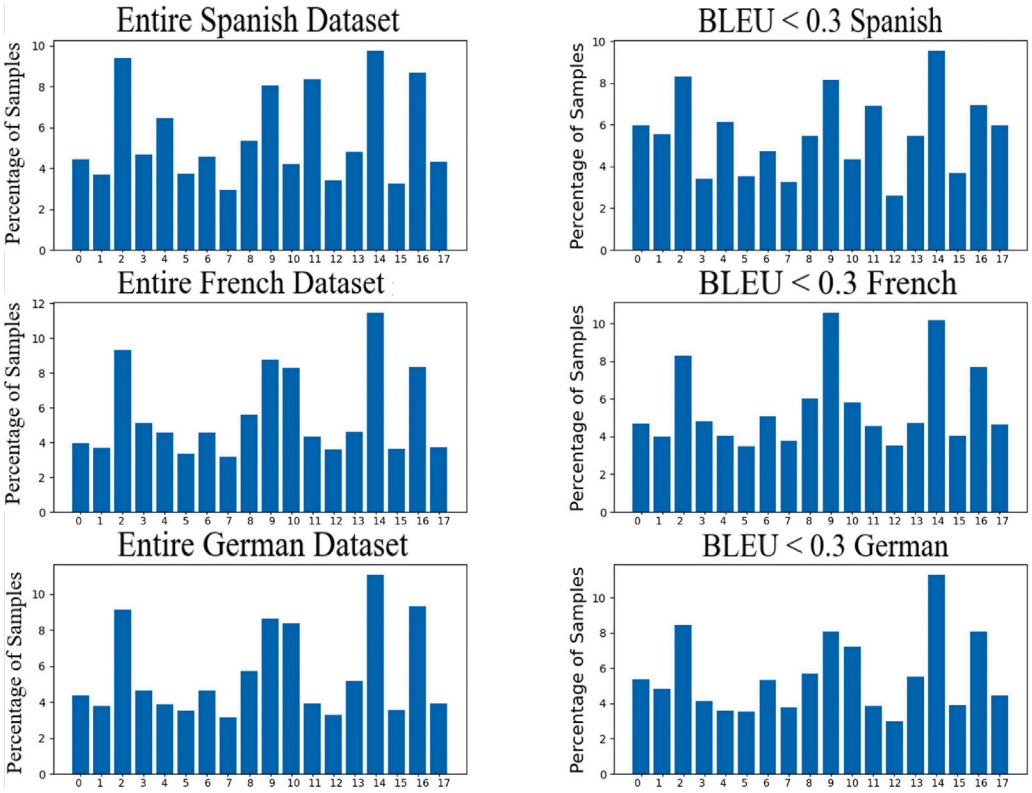


Fig. 2. Distribution of test set samples per class before and after preprocessing in Spanish, French, German.

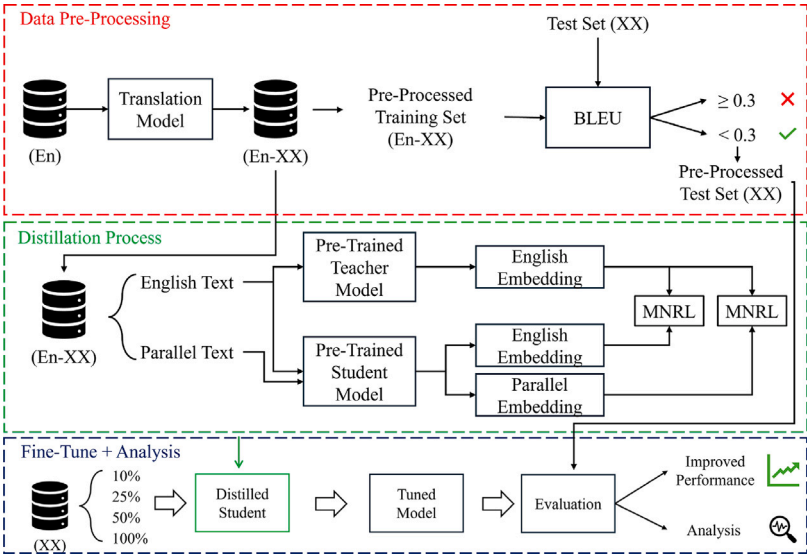


Fig. 3. Our pipeline consists of three main stages. **Data Preprocessing:** Creation of an unbiased dataset. **Distillation Process:** Alignment of teacher and student embeddings using MNRL. **Fine-Tuning and Analysis:** Training and evaluation of the generated student models.

of the current state-of-the-art open-source neural machine translation models, making it particularly suitable for high-quality multilingual translation tasks.

Moreover, to assess the robustness of our approach across various amounts of available data, we conducted experiments with different subsets of the training data. Specifically, we evaluated the performance of the model using 10%, 25%, 50%, and 100% of the dataset. This allowed us to measure the scalability and effectiveness of the proposed method under both low-resource and full-resource conditions.

Finally, to better understand the structural relationship between the teacher and student models post-distillation, we performed Representational Similarity Analysis. This method allowed us to quantitatively compare the embeddings produced by the teacher and student models. RSA computes the pairwise cosine similarities between token embeddings on a shared test set and then measures the correlation between the resulting similarity matrices of the teacher and student models [41], providing a quantitative assessment of their alignment.

3.3. Models

We conducted our extended experimental phase by employing Transformers models for both teachers and student. The basic idea behind Transformers is to leverage a self-attention mechanism that allows each token in the input sequence to attend to all other tokens, capturing long-range dependencies more effectively than traditional recurrent networks. This architecture, introduced in [42], enables the parallelization of computations and scales efficiently with large datasets. These models have achieved widespread adoption due to their scalability, allowing them to be pre-trained on vast corpora and then fine-tuned with minimal data on specific downstream domains and tasks. This paradigm has established Transformers as the state-of-the-art approach not only in NLP but also across various other fields in recent years. BERT (Bidirectional Encoder Representations from Transformers) [6] is a notable example of this architecture. We therefore decided to use BERT-based models for two key factors: first, they offer a cost-effective solution in terms of computational resources; second, both domain-specific adaptations (BioBERT [5]) and multilingual variants (mBERT [6]) are well-established in the literature, all sharing a consistent core architecture. This consistency allowed us to avoid biases introduced by using different model architectures. The former was further pre-trained from the original BERT model on PubMed abstracts and PMC full-text articles, and has demonstrated superior performance on several biomedical text mining tasks with respect to the original, general-domain models [5,43]. The latter was trained on the top 104 languages with the largest Wikipedias using both masked language modeling and next sentence prediction objectives similar to the original English BERT, and has been shown effective in handling inputs from multiple languages [44].

3.4. Experiments

To answer the research questions presented in the introduction, we conducted a series of experiments designed to evaluate the effectiveness of our approach in domain and task knowledge transfer, as well as the impact of the fine-tuning data size:

- To evaluate the potential of our approach for transferring domain knowledge between languages (**RQ1**), we utilized the pre-trained BioBERT as the teacher model (B_p) and mBERT as the student model for the target languages.
- To assess the capability of transferring task knowledge between languages (**RQ2**) and to determine the benefits of having a model pre-trained on the domain (**RQ3**), we fine-tuned both BioBERT and mBERT models on the task. These fine-tuned models (B_{ft} and mB_{ft}, respectively) were then used as teachers for the non-tuned mBERT in the target languages.
- To examine the impact of the amount of tuning data (**RQ4**), we fine-tuned all models using different volumes of data, specifically 10%, 25%, 50%, and 100% of the original training dataset in the available languages.

The fine-tuning process encompassed all four languages (English, Spanish, French, and German), while the cross-lingual KD phases involved transferring knowledge from English to Spanish, French, or German. The students were then compared to their counterparts tuned without passing through the KD process ($noKD$).

3.4.1. Framework analysis

To further evaluate the changes in our models (**RQ5**) and to examine the effectiveness of our approach, we conducted additional analyses. Specifically, we aimed to assess how closely the distilled models resemble their respective teachers, both before and after fine-tuning.

First, we employed the Representational Similarity Analysis technique, widely used in recent literature [39,45,46], to quantify changes in the representations of BERT-like models following training. Using this technique, we extracted the *[CLS]* token representations of $N = 4300$ English test samples from the teachers and the translated test samples from the students (both before and after distillation). For each model, we computed the $N \times N$ pairwise cosine similarity matrix and flattened the upper triangular part to obtain an array. Finally, following previous works, we computed the Pearson correlation between the representations of the teachers and students.

In the second analysis, we compared the Euclidean distances of the confusion matrices obtained from the test samples in the target languages. We then created a heatmap to visualize how our methodology influences the distances between the different student models. Specifically, we aimed to determine if using KD results in more similar student models compared to not using KD.

4. Results and discussions

Table 3 summarizes the results of the experiments conducted to address the research questions posed in this study. It shows the performance of the models distilled from three different teachers: the pre-trained BioBERT (B_p), the fine-tuned BioBERT (B_{ft}), and the fine-tuned mBERT (mB_{ft}). The table includes results for all three target languages and compares the performance of the distilled models with their non-distilled ($noKD$) counterparts. Notably, the proposed approach consistently enhanced downstream performance across all teacher models, achieving improvements of up to +12.89% in the F_1-score, with minor degradations as low as −1.77%. The average performance increase (+3.81%) was substantially higher in magnitude compared to the average decrease (−0.89%).

Specifically, our approach successfully transferred both domain knowledge from the pre-trained teacher and task-specific knowledge acquired during fine-tuning from the two teachers: the domain-adapted BioBERT and the general-domain mBERT, which answer to **RQ1** and **RQ2**. In particular, among the three languages, there is no clear distinction in performance when distilling task knowledge from the domain-adapted model versus the general-domain model (**RQ3**). Conversely, using the non-tuned BioBERT as the teacher often led to better improvements compared to its fine-tuned version, with an average increase of approximately +0.70% in F_1-Score. These findings underscore the value of domain-specific pre-training. However, as mBERT yielded comparable performance improvements when used as a teacher model, it presents a more cost-effective alternative to developing a domain-adapted model from scratch.

However, the benefits of our approach are not uniformly distributed across different experimental setups. The performance gains achieved through knowledge distillation diminish as the volume of tuning data increases (**RQ4**). Specifically, the average increase in F_1-Score was +7.95% when using 10% of the dataset, but this improvement diminished to approximately +0.45% when using 100% of the dataset. Intermediate results were around +1.72% and +1.05% for 25% and 50% of the dataset, respectively. This trend aligns with expectations, as it mirrors the performance of the domain-adapted model and mBERT (across all four languages) when fine-tuned without KD, as illustrated in Fig. 4. In this figure, the F_1-Score gap of about +10% between BioBERT and mBERT when tuned on 10% of the English dataset is partially reduced to +5% when trained on the entire dataset.

To further evaluate our approach, we compared it with the state-of-the-art multilingual method proposed by Reimers and Gurevych [27]. Their method, unlike most existing approaches, avoids the use of knowledge distillation during fine-tuning, enabling a broader scope of experimentation (e.g., RQ1). Given its close alignment with our research goals, we chose to reimplement and benchmark it directly against our own approach. Specifically, we reproduced their method using the Mean Squared Error loss function and present the results in Fig. 5. The figure compares the performance of each teacher model

Table 3
F_1-Scores (%) for Spanish, French, and German at different dataset sizes, for models without knowledge distillation ($noKD$), for models with pre-trained BioBERT as the teacher (B_p), for models with fine-tuned BioBERT on the task as the teacher (B_{ft}), and for models with fine-tuned mBERT on the task as the teacher (mB_{ft}). Scores in red indicate worse performance compared to $noKD$, scores in green indicate better performance, and scores followed by an asterisk (*) indicate the best teacher for a specific language and tuning data size.

Data size	Spanish - F_1-Score				French - F_1-Score				German - F_1-Score			
	$noKD$	B_p	B_{ft}	mB_{ft}	$noKD$	B_p	B_{ft}	mB_{ft}	$noKD$	B_p	B_{ft}	mB_{ft}
10%	73.46	+3.44	+4.43	+4.65*	63.73	+12.87	+12.17	+12.89*	64.49	+8.81*	+4.61	+7.68
25%	80.46	+0.11	+1.14*	+0.57	78.21	+5.59*	+4.07	+4.49	78.34	+0.30*	+0.17	−1.00
50%	86.85	−0.03	−0.18	+1.45*	86.43	+2.27*	−0.04	+0.00	81.05	+2.45*	+1.93	+1.60
100%	90.24	+0.63*	−1.71	−1.77	89.42	+1.56	+1.76*	+0.43	87.45	+2.18	+2.48*	−1.49

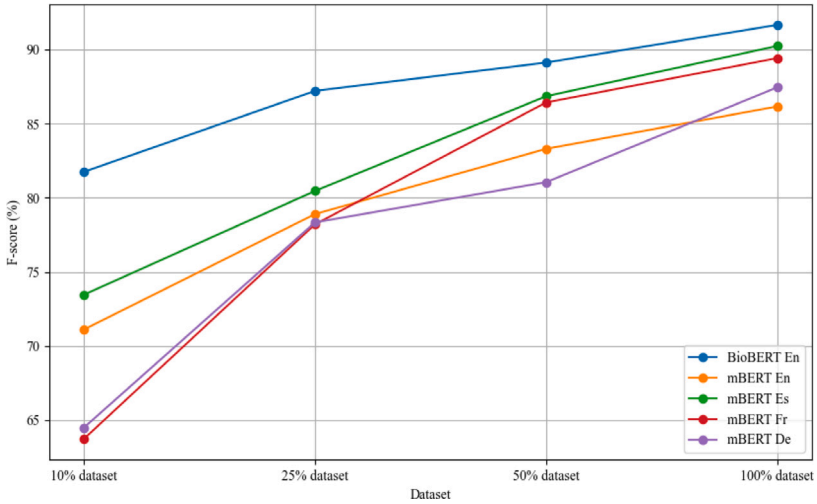


Fig. 4. F_1-Score (%) results for BioBERT and mBERT fine-tuned models.

across multiple languages to the noKD baseline. Notably, MNRL outperforms MSE, particularly when data is scarce, delivering superior results in 32 out of the 36 evaluated cases.

When applied to machine-translated training data, the MSE loss function often underperforms relative to the noKD baseline, as it is highly sensitive to noise introduced by automatic translations. This sensitivity can lead to performance degradation below that of the baseline [47,48]. In contrast, MNRL demonstrates greater robustness to data noise, enabling it to better manage the imperfections inherent in automatic translations and improve overall performance [49].

4.1. Framework analysis

To further analyze and understand the behavior of our framework, we compared the performance of students and teachers both before and after fine-tuning (**RQ5**).

Using the RSA technique to compare teacher and student models before and after knowledge distillation (Fig. 6) revealed a higher correlation between the teacher and the distilled student. The observed increase in correlation was consistent across languages and teacher models, underscoring the effectiveness of our approach in achieving better alignment. The enhanced similarity reflects successful knowledge transfer, corroborating the improved F1-scores reported in Table 3. Additionally, it was observed that the students exhibited significantly higher correlations with the non-tuned teacher models, both before and after KD. This outcome likely stems from the significant alterations to the model's layers, particularly the final layers, during fine-tuning [50–52]. Fine-tuning causes models to specialize in capturing task-specific information, at the expense of the broader knowledge acquired during pre-training, a phenomenon known as catastrophic forgetting, thus focusing on task-specific terminology [53,54]. Indeed, while BioBERT and mBERT were pre-trained on different corpora, they went through the same training objectives, thus resulting in more similar models.

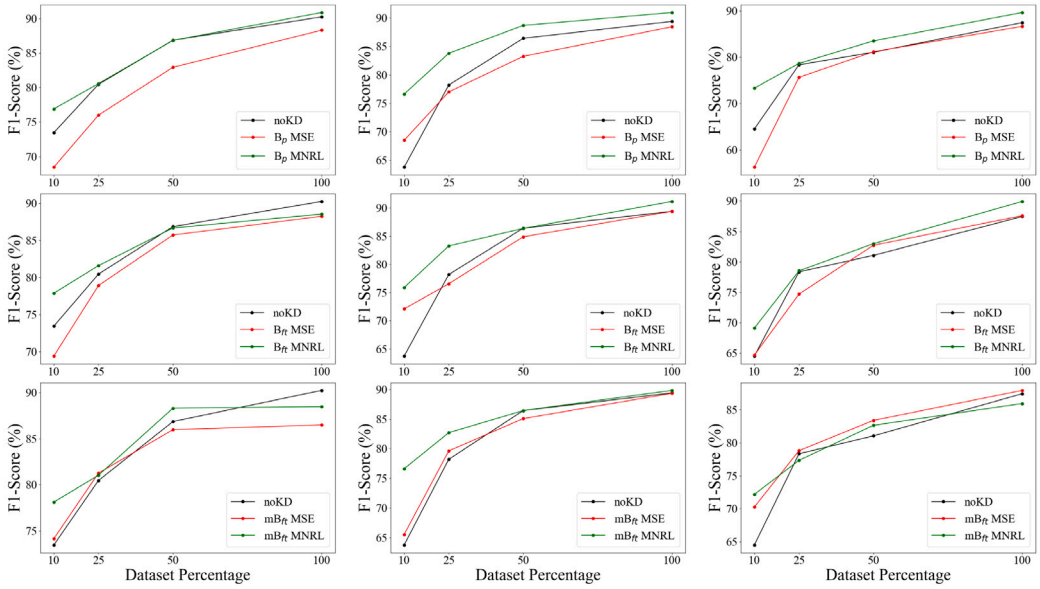
Moreover, to assess the resemblance of the students to the teachers after fine-tuning, for each target language we extracted the confusion matrices of the four student models and computed the Euclidean distance for each pair. Fig. 7 visualizes the comparisons through heatmaps, where each cell represents the distance between the confusion matrices of the model on the y-axis and the model on the x-axis. This analysis, as can be observed in the figure, revealed that regardless of the employed teacher, the distilled models are more similar to each other than to the non-distilled model. This result was consistent across all the three target languages.

5. Conclusion

This study introduces a cross-lingual knowledge distillation approach using Sentence Transformers to transfer domain- and task-specific knowledge from English to other languages (Spanish, French, and German) with limited domain-specific and task-specific training data. The proposed approach demonstrated improved downstream performance, particularly in data-constrained scenarios, highlighting its effectiveness across various target languages and its potential to reduce annotation and training costs.

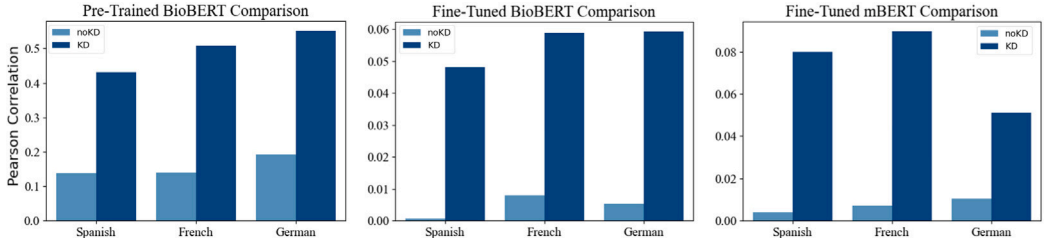
Comprehensive analyses addressed key research questions, offering insights into the effectiveness of the proposed approach, the influence of varying data sizes, and the differences in internal representations between teacher and student models. The findings underscore the scalability and adaptability of the proposed method across languages, illustrating the progressive alignment of the student model with the teacher model.

Given the limitations imposed by the token count during the distillation process, future work could investigate the impact of varying token counts on the model's performance. Future work will also explore applying this technique to monolingual student models. As monolingual models often surpass their multilingual counterparts in performance, transferring knowledge to such models could yield further improvements in downstream tasks.



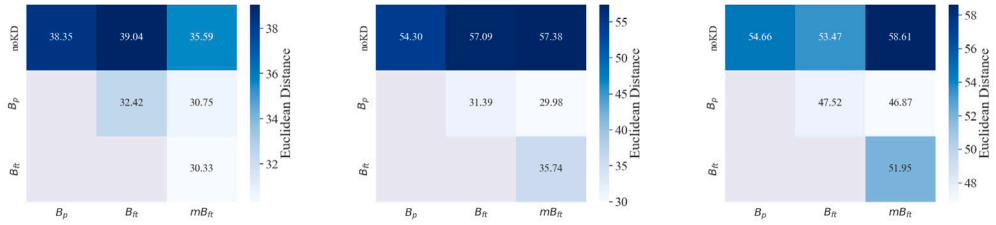
(a) Spanish (b) French (c) German

Fig. 5. F1-Scores using MSE and MNRL for three different teachers (B_p, B_{ft}, and mB_{ft}) with respect to **noKD** baseline across varying dataset percentages for (a) Spanish, (b) French, and (c) German. This comparison highlights the relative performance improvement of each teacher model over the baseline.



(a) B_p (b) B_{ft} (c) mB_{ft}

Fig. 6. RSA comparisons between teacher and student models before and after the Knowledge Distillation process, using (a) B_p, (b) B_{ft}, and (c) mB_{ft} as the teacher models.



(a) Spanish (b) French (c) German

Fig. 7. Heatmaps to compare the Euclidean distance between confusion matrices for each pair of models for (a) Spanish, (b) French, (c) German.

CRediT authorship contribution statement

Ruben Piperno: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation. **Luca Bacco:** Writing – original draft, Visualization, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Felice Dell'Orletta:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Mario Merone:** Writing – review & editing, Supervision, Project administration, Methodology, Data curation. **Leandro Pecchia:** Writing – review & editing, Visualization, Supervision, Project administration.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to enhance the readability of the manuscript, based on text originally written by the authors. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table A.4
Class names and their corresponding versions.

Index	Class name	Version
0	Brain, spinal cord, and nerve disorders	Patient
1	Cardiovascular disorders	Professional
2	Children's health issues	Patient
3	Eye disorders	Both
4	Gastrointestinal disorders	Professional
5	Genitourinary disorders	Professional
6	Gynecology and obstetrics	Professional
7	Hematology and oncology	Professional
8	Infections	Patient
9	Infectious diseases	Professional
10	Injuries and poisoning	Patient
11	Injuries; Poisoning	Professional
12	Musculoskeletal and connective tissue disorders	Professional
13	Neurologic disorders	Professional
14	Pediatrics	Professional
15	Pulmonary disorders	Professional
16	Special subjects	Both
17	Women's health issues	Patient

Appendix. Class information

See Table A.4.

Data availability

Data is publicly available. We report in the manuscript the source of the data employed in the study.

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